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Multi-channel EEG Signal Analysis for Mental Arithmetic Task Classification using Hybrid CNN-LSTM Model

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***Abstract*— The mental effort needed to complete the activities is referred to as cognitive workload. Mental arithmetic tasks (MAT) are engaging in various cognitive processes, including decision-making, memory recall, problem solving, fact retrieval, etc. Solving a MAT requires more than one brain activity and is a crucial component in many fields, such as industrial applications, clinical medicine, and system design. An EEG (electroencephalogram) is an effective tool for detecting patterns and variations in the brain's electrical activity. Convolutional Neural Network (CNN) have been widely applied for automatic feature extraction from raw EEG signals for different deep learning (DL) models. Due to the capability of temporal dependency representation, Recurrent Neural Networks (RNN) can be considered suitable for the time series analysis of EEG signals. Considering the advantages of DL models, a hybrid CNN-LSTM model is developed for the categorization of EEG signals obtained before and during the MAT. The proposed model is assessed by utilizing the open access MAT EEG database and is obtained during the performance of serial subtraction of two numbers. The proposed approach is executed utilizing MATLAB R2023b and the performance metrics such as accuracy (99.10%), sensitivity (96.05%), specificity (99.72%), precision (98.64%), MCC (96.80%), kappa (99.09%) and F1-score (97.33%) are obtained respectively and are compared with the existing methodologies.**

Keywords*— *Cognitive workload, EEG, mental arithmetic task, deep learning.

I. Introduction

Mental tasks, such as reality retrieval, memory, sequencing, and decision-making, are essential cognitive functions. Mental activities are essential for improving cognitive flexibility and critical thinking skills. Monitoring these activities through EEG signals helps to understand how the brain responds to such tasks. Research has highlighted the importance of analysing brain activity for early detection of various cognitive and mental disorders. This includes syndromes like autism, dyscalculia, attention deficit hyperactivity disorder[1]. The study of electrical brain activity during mental tasks helps in understanding and addressing these conditions.

Fig. 1. Performance of Mental Arithmetic Task

The brain cortex comprises four major lobes frontal, parietal, temporal, and occipital[2]. These lobes coordinate and facilitate many cognitive operations, including numerical processing, memory retrieval, and construction of problem-solving strategies. The frontal lobe involves in decision-making, working memory, and problem-solving, while the parietal lobe deals with sensory processing and numerical calculations and retrieval of information from facts. The auditory system and verbal memory may be grouped with the temporal lobe, whereas the occipital lobe is related to visual perception and the recognition of numerical symbols.

The EEG is a non-invasive method widely utilized for analysing cognitive performance across various brain states[3]. It is particularly effective in assessing brain activity during cognitive tasks. Electrode placements are determined utilizing the International 10/20 system, which aligns electrode positions with the cerebral cortex structure for accurate data acquisition. EEG signals are beneficial for testing functional aspects of the human brain and refer to a change in powering patterns of brainwaves, often in a MAT. Its biggest advantage is speed, as it is able to detect complex brain activity within fractions of a second post-trigger. EEG is

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employed for various purposes in both clinical and cognitive settings. It is utilized to monitor brain activity and to find abnormal brain wave activity that can assist in the assessment of neurological disorders such as epilepsy. EEG may also be used when monitoring multiple sleep stages in sleep studies to assess sleep disorders such as insomnia and obstructive sleep apnea. In cognitive research, EEG is used to measure brain activity for a variety of tasks such as learning, memory, problem-solving and brain activity associated with attention.

Several authors have examined EEG signals during MATs to study brain activity in both resting and active states. Binish Fatimah et. al.,[1] though Fourier decomposition method is used for decomposing the EEG into multiple sub-bands and features were extracted from each sub-band. The MATs were categorized using machine learning (ML) method based on the extracted features. Binish Fatimah et. al., presented a ML classifier methodology for the detection of MAT difficulty level [2]. The digital filters were utilized and decomposed the raw data into various frequency rhythms and extracted the features. The extracted features were utilized and classified the data into rest and active states using various ML classifiers. Naqib Sad Pathan et.al., proposed a SVM classifier to categorize the MAT [5]. The signals were divided into discrete frequency sub-bands utilizing the discrete wavelet transform in which the features were statistically assessed. Qiang Wang et.al.[6] performed multifractal analysis of EEG data using a method known as Generalized Higuchi Fractal Dimension Spectrum. In MAT classification, the algorithms applied can be subject-dependent and may operate in either a single-channel or multi-channel configuration. The proposed method demonstrated an accuracy of 84.15% using single-channel algorithms, which further improved to 97.87% with multi-channel subject-dependent algorithms. Biswarup Ganguly et.al., proposed a stacked LSTM model [7] to recognize and categorize MATs with less number of electrodes. Window segmentation was implemented to enhance data augmentation, resulting an acc of 91.67%. Celia Shahnaz et.al., utilized Perceptual Wavelet Packet Transform [8] to separate EEG signals into multiple frequency bands. Teager energy was computed for the wavelet packet coefficients and extracted the features. These features were utilized and categorized using SVM. Rifai Chai et.al. suggested an artificial neural network for categorization that optimizes genetic algorithms [9]. The data is collected from subjects using non-hair area electrodes on the prefrontal cortex. Hilbert-Huang Transform energy method were utilized for feature extraction and achieved 73% of accuracy for mental tasks vs. baseline conditions. Hyeon Kyu Lee et.al. proposed a CNN to classify the motor imagery [10]. Time-frequency images of EEG signals were generated using the Continuous Wavelet Transform and achieved 97.61% of accuracy. Abhishek Varshney et al. [11] investigated classification of good and bad MA calculation, and classification before and during MA calculation, using RNN classifiers. The RNN models achieved accuracy values exceeding 99% in all cases. Author in [12] examines the mental workload in real time applications, such as driving under real traffic conditions. Classified mental workload levels in real-time using ML methods. Tahemineh Azizi [13] begins by outlining the algorithms used in signal processing. Firstly divides the input signal into overlapping segments using the STFT, and then applies FT to each segment. Nan Ying Liang introduced in [14] a new classification algorithm called the Extreme Learning Machine that reduces training time with no regard for the loss of acc. In this model, feature extraction was made on raw data and then classifiers were trained based on feature vectors. Then, the classifiers were tested using the unseen data for classification acc evaluation. Abadi Hoda Edris[15] extracted geometric features such as short distance, length, angle and slope. The most relevant geometric features are selected for dimensionality reduction and improvement of the ML model's efficiency. The selected features are fed into a ML classifier to detect and classify MATs.

Despite advancements, EEG-based mental arithmetic classification faces several limitations. Subject-dependent algorithms may not generalize well across individuals, and high-dimensional feature sets increase computational complexity. Many methods require extensive preprocessing, making real-time applications challenging. ML models demand large datasets for effective training, which are not always available[16]. Additionally, signal quality is affected by noise, artifacts, and electrode placement, reducing model reliability.

This paper presents the following key contributions:

(1) EEG signals are pre-processed with a band-pass Butterworth filter of order 2 followed by Z-score normalization to enhance the quality of data.

(2) CNN and LSTM networks are combined to automatically extract spatial and temporal features from EEG signals.

(3) he features extracted from the CNN-LSTM fusion model are used for classifying EEG data into two classes: before and during the task.

(4) The model proposed by this method was evaluated using the MIT Physionet EEG database and achieved a much better performance than already-existing methods.

With this paper going ahead, Section II serves as a brief introduction to the project that includes a block diagram, dataset details, and methodology. Section III presents a detailed analysis of the simulation results, evaluating the performance of the proposed method using various metrics and comparing it with existing methods. Section IV summarizes the key outcomes of the study and future work.

II. Proposed Methodology

MAT involve solving math problems in the mind without external tools engaging cognitive processes like attention, working memory, and problem solving. EEG can capture brainwave patterns which reflect cognitive engagement, focus and mental workload during these tasks[17]. The proposed

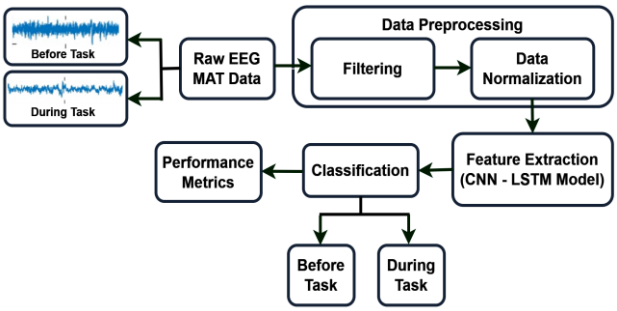

Fig. 2. Block Diagram of the MAT Classification Model

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hybrid CNN-LSTM method includes three stages, data pre-processing, feature extraction and classification. The block diagram of the proposed method is presented in Fig. 2.

This Fig. 2. illustrates the process of analysing EEG data to classify brain activity before and during a MAT task. The raw data signals are recorded from individuals, which include the brainwave signals recorded under two conditions, before and during the task. The first stage is data preprocessing, which consists of two stages, filtering and normalization. The Butterworth bandpass filter removes noise and unwanted frequency components. Normalization rescales the data so that the comparison among the signals could easily be processed in a uniform way by deep-learning algorithms[18]. After pre-processing, feature extraction takes place using the hybrid deep learning model that integrates CNN and LSTM (Long Short-Term Memory). While CNN acts on spatial features[19], LSTM works well for sequential data. This combination extracts the characteristics from EEG data by temporal and spatial segmentation, allowing the model to better distinguish brain states before and during the task. The extracted features are utilized to categorize the EEG signals into two distinct groups: Before Task (BT) and During Task (DT). This performance classification is evaluated using metrics accuracy (acc), sensitivity (sen), specificity (spe), precision (pre), Matthews Correlation Coefficient (MCC), kappa (ka) and F1-score (F1_s). The analysis helps in understanding the changes in mental states during task performance.

A. Database Description

In this study, an open access dataset from MIT PhysioNet was utilized[20]. EEG recordings were collected from individuals both before and during MAT and are included in the database. Silver or silver chloride electrodes were arranged symmetrically on the scalp and the Neurocom 23-channel system (Ukraine, XAI-MEDICA) was utilized to record the EEG data.

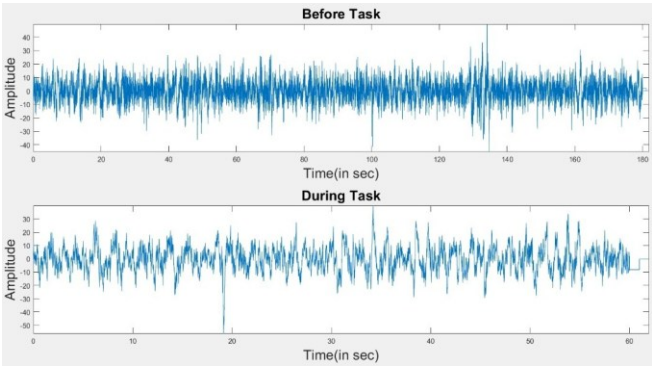

Fig. 3. Waveforms of Before and During the performance of MAT

The impedance between electrodes was maintained less than 5 kΩ by referencing each electrode to the ear electrodes. EEG signals were filtered with a high-pass at 0.5 Hz and a low-pass at 45 Hz, and sampled at 500 Hz per channel. To eliminate power line noise, a 50 Hz notch filter was utilized.The amplification tract was lasted for 0.3 seconds. There are 36 subjects in total. These EEG recordings are artifact-free recordings of 180 sec before task and 60 sec during task. Two numbers were serially subtracted as the arithmetic task. Every trial began with an oral discussion of two-digit (subtrahend) and four-digit (minuend) integers, such as 3141 and 42. The before task data includes 36 subjects, however 33 subjects have 91000 samples but the remaining three subjects has 94000, 85000 and 40000 samples for 182s duration each, while the during task data has 31000 samples for all the 36 subjects of 60s duration each. The EEG segments are separated into non-overlapping chunks to improve the data available for training and testing. A window size of 35s (35*500=17500 samples) was considered to ensure that the signals are the same length before and during the task A total of 4452 frames are considered for analysis (BT-3696 frames and DT-756 frames). Z-score normalization is utilized to standardize the data distribution effectively and expressed in equation (1) as

$$\text{Z Score} = \frac{\text{(data - mean of the data)}}{\text{std deviation of the data}} \tag{1}$$

B. CNN-LSTM Model

Fig. 4. shows the proposed hybrid CNN-LSTM model's block diagram. For the purpose of classifying mental arithmetic tasks with EEG, a hybrid Convolutional Neural Network-Long Short-Term Memory (CNN-LSTM) model was employed. This model was selected due to the advantage of combining both CNN and LSTM in the learning process of the spatial-temporal representations of EEG data. The CNN model is well-suited for capturing spatial dependencies across EEG channels, while the LSTM model represents the temporal dynamics present in the EEG data due to sequential neural activity. The combination gives strength to the feature extraction and classification, especially in relation to brain signals associated with performing cognitive tasks. The CNN architecture has several layers that work together to process input. The convolutional layer (CL) utilizes a filter to extract features, generating feature maps. The dimension of the feature maps is minimized by utilizing a pooling layer (PL) to improve the process more efficiently. Pooling is categorised into two types. First, Max pooling selects the largest value while, average polling gives the average value from feature map. The neurons are connected between various layers through the use of a fully connected layer (FL). It includes the neurons, along with their associated weights and biases. In proposed model, the CNN block includes three CL, each with a filter size of three. The number of filters is fixed to 128, 256, and 128, with padding set to "same" and a stride of 2 for all layers. A ReLU activation layer is utilized to add non-linearity and improves feature extraction after each CL. A global average PL is applied to minimize the number of values and further optimize the network. This pooling method computes the average of each feature map, providing a compact representation of the input data while retaining its essential features.

LSTM architecture consists of an input, hidden state, cell state, gates, and output. At each time step, it processes an input, updates the hidden state and maintains long-term memory via the cell state. Three gates regulate information flow, the Forget Gate selectively erases past data, the Input Gate adds or discards new information, and the Output Gate decides which aspects of the cell state to focus on or ignore when computing the output. Here, LSTM block consists of an LSTM and a dropout layers. Two LSTM layers were considered with hidden units of 64 and 128. The output mode for first layer is taken as sequence and for second layer is considered as last. After each LSTM layer, dropout layer of

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50 is utilized. Concatenation layer combines the outputs of both CNN and LSTM blocks.

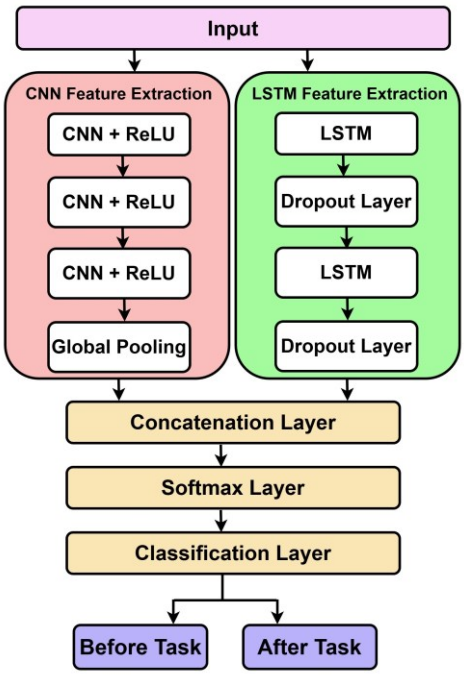

Fig. 4. Block Diagram of the proposed hybrid CNN-LSTM Model

The concatenation layer's output is applied to the two FL with ReLU activation and then it is given to the softmax layer. The classification layer is utilized to classify the EEG data into Before task and During task.

III. Simulation Results

The hybrid CNN-LSTM model's performance in MAT database classification is evaluated using acc, sen, spe, pre, F1_s, MCC and ka metrics. The definitions and mathematical relations are derived above.

Accuracy: It evaluates the ratio of instances that are correctly classified.

$$\text{Acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \times 100\% \tag{2}$$

Sensitivity: It calculates the proportion of true cases to all positive cases.

$$\text{Sen} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\% \tag{3}$$

Specificity: The proportion of actual negative values that are correctly classifies as negative.

$$\text{Spe} = \frac{\text{TN}}{\text{TN} + \text{FP}} \times 100\% \tag{4}$$

Precision: The acc of positive classifications, measured as the ratio of TP predictions among all predicted positives.

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\% \tag{5}$$

F1 score: The harmonic average of sen and pre, balancing both metrics to measure the model acc, especially in imbalanced datasets.

$$\text{F1_S} = 2 \times \frac{\text{Pre} \times \text{Sen}}{\text{Pre} + \text{Sen}} \times 100\% \tag{6}$$

MCC: It determines the correlation between actual and predicted labels to evaluate the validity of binary classifications.

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})\,(\text{TP} + \text{FN})\,(\text{TN} + \text{FP})\,(\text{TN} + \text{FN})}} \times 100\% \tag{7}$$

Kappa: It utilized to assess the level of agreement between two raters.

$$\text{Ka} = \frac{\text{Acc} - \text{Pe}}{1 - \text{Pe}} \times 100\% \tag{8}$$

In above formulas, TP (True Positive) and TN (True Negative) correctly predicted positive and negative outcomes. FP (False Positive) and FN (False Negative) incorrectly predicted positive and negative outcomes. The proportion of agreement that would result from raters classifying at random is known as Expected agreement (Pe).

A. Training Process

For CNN-LSTM model training, validation, and testing, the database was split into 80%, 10%, and 10% respectively. A mini-batch size of 64 is utilized for the training dataset. The The cost function is optimized by utilizing the Adam optimizer. The training performance of a model over five epochs. Random shuffling was used after each epoch to reduce bias during training and iteration and help in faster training convergence The training was completed efficiently in 58 seconds ,with a total of 275 iterations (55 per epoch). The model effectively converged with a piecewise learning rate, achieving low loss and high acc.

B. Model Evaluation and Discussion

Table I presents the performance evaluation of hybrid CNN-LSTM method for classify MAT data with different overlap sizes (25%, 50%, 75%, and Non-Overlap).

TABLE I. Evaluation of Hybrid CNN-LSTM Model metrices with Different Overlap Sizes

Overlap Size	25%	50%	75%	Non-Overlap
Acc	0.946	0.971	0.954	**0.991**
Sen	0.848	0.911	0.877	**0.960**
Spec	0.971	0.985	0.981	**0.997**
Pre	0.879	0.903	0.944	**0.986**
F1_s	0.863	0.920	0.909	**0.973**
MCC	0.831	0.903	0.880	**0.968**
Ka	0.944	0.971	0.952	**0.990**

From the results, observed that all of the performance metrics improve when the overlap size is increased from 25% to 50%. When the overlap increases to 75%, some metrics,

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such as sensitivity and F1 score, slightly decrease compared to 50% overlap due to redundancy. The non-overlapping situation provides highest values for all metrics, indicating that the use of distinct EEG segments improves the acc and reliability of MATs classification.

Table II presents the performance of the model for Non-Overlap classification across different window sizes (WS) ranging from 5 to 40 seconds. Metric values are improved with increasing window size. Acc and sen are lower when the window size is 5 seconds, but as the window size grows, both metrics improved significantly. At 30 seconds, acc reaches 96.13%, with sensitivity and precision also achieved highest levels.

TABLE II. ANALYSIS OF THE HYBRID CNN-LSTM MODEL VARYING DIFFERENT WINDOW SIZES

WS (sec)	Acc (%)	Sen (%)	Spe (%)	Pre (%)	F1_S (%)	MCC (%)	Ka (%)
5	90.01	79.95	93.23	79.04	79.49	72.90	89.03
10	92.57	81.38	96.46	88.88	84.97	80.19	92.03
15	92.71	85.00	95.30	85.85	85.42	80.57	92.18
20	95.42	87.26	97.95	92.96	90.02	87.13	95.52
25	95.97	86.36	98.33	92.68	89.41	87.01	95.81
30	96.13	89.61	98.41	95.17	92.30	89.80	95.98
35	**99.10**	**96.05**	**99.72**	**98.64**	**97.33**	**96.80**	**99.09**
40	97.06	91.30	98.36	92.64	91.97	90.18	96.98

The model achieved better in Non-Overlap classification for the 35-second WS across all metrics. While performance slightly drops at 40 seconds, larger window sizes generally improve model performance, with the 35-second window providing the best results.

Table III shows a comparative performance evaluation of four models CNN in series, LSTM in series, CNN-LSTM in series, and the Hybrid CNN-LSTM model evaluated across multiple performance metrics. So far, the Hybrid Model has exhibited the best classification performance with the highest values on acc (99.10), pre (98.84), and F1_s (97.33), showing a balance between precision and reliable prediction. The CNN and LSTM models seem to perform well, but CNN-LSTM and Hybrid Model thus have additional benefits that are a combination of the convolutional and recurrent architecture.

TABLE III. MODEL PERFORMANCE METRICS

Model	CNN	LSTM	CNN-LSTM	Fusion Model
Acc	0.941	0.955	0.975	**0.991**
Sen	0.829	0.855	0.888	**0.960**
Spe	0.969	0.975	0.994	**0.997**
Pre	0.869	0.878	0.972	**0.986**
F1_s	0.848	0.866	0.929	**0.973**
MCC	0.813	0.839	0.915	**0.968**
Ka	0.938	0.953	0.974	**0.990**

Table IV presents a comparative performance analysis of the proposed model and the model accuracies attained by various authors. This include a number of previous studies, their performances are analyzed and they are compared to the proposed model. The accuracies of earlier models range from a low of 51.00 to a high of 98.60. The Extreme Learning Machine Model of Liang N et al. and that of Del R Millan et al., in real-time mental workload identification, perform more poorly at 51.00 and 70.00, respectively. Moreover, it is on the other hand that advanced strategies are much greater. Among them include the real-time MAT recognition of Wang et al. (97.10) and Fourier decomposition method of Binish Fatimah (98.60). The proposed Hybrid CNN-LSTM model achieves an acc of 99.10, stands out with the highest acc in the table and far exceeds all these techniques.

TABLE IV. PERFORMANCE COMPARISION OF PROPOSED HYBRID CNN-LSTM MODEL WITH EXISTING MODELS

S. No	Author	Accuracy (%)
1	Biswarup Ganguly [7]	93.59
2	Ramaswamy [21]	94.40
3	Del R Millan et al [22]	70.00
4	Wang et al [6]	97.10
5	Anderson et al [23]	91.40
6	Liang n et al [14]	51.00
7	Binish Fatimah [1]	98.60
8	**Proposed CNN-LSTM Hybrid model**	**99.10**

IV. CONCLUSION

The classification of MAT based on EEG signals is much needed in cognitive neuroscience and BCI applications[24]. EEG signals are non-invasive and high in temporal resolution, present challenges due to their complex and non-stationary nature. To overcome the limitations, an integrated CNN-LSTM model built on CNNs and LSTMs has been proposed. The spatial features are extracted by CNNs and LSTMs for temporal sequence modeling, thus enhancing the model's performance with regard to classification accuracy. The model achieved remarkable accuracy of 99.31, along with various performance metrics such as sensitivity (96.05), specificity (99.72), precision (98.64), Matthews Correlation Coefficient (MCC) (96.80), recall (96.05), kappa (99.09) and F1-score (97.33). This indicates that the proposed method has been effective in accurately categorizing EEG signals during MAT. For future research, efforts could focus on enhancing the model's ability to generalize across diverse participants and varying conditions by improving its performance on larger and more complex EEG datasets. Additionally, the incorporation of transformer-based architectures and attention mechanisms could be explored to further strengthen feature extraction and overall model capabilities.

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